ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

INTRODUCTION TO N-GRAM LANGUAGE MODELS

UC San Diego

COMPUTER SCIENCE & ENGINEERING

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N-Gram Models

Use chain rule to generate words left-to-right

$$P(w_1 \ldots w_n) = \prod_i P(w_i | w_1 \ldots w_{i-1})$$

Conditioning on the entire left context is difficult

P(??? | Turn to page 134 and look at the picture of the)

N-gram models make a Markov assumption

$$P(w_1 \ldots w_n) = \prod_i P(w_i | w_{i-k} \ldots w_{i-1})$$

$$P(\text{please close the door}) =$$

$$P(\text{please}|\text{START})P(\text{close}|\text{please}) \ldots P(\text{STOP}|door)$$

How do we know P(w | history)?

- Use statistics from data (i.e. count n-grams in large corpus)
- For example, if we want to estimate P(door | the):

Training Counts

198015222 the first
194623024 the same
168504105 the following
158562063 the world
…
14112454 the door

23135851162 the *

$$\hat{P}(\text{door}|\text{the}) = \frac{14112454}{23135851162}$$
$$= 0.0006$$

- This is the *maximum likelihood* estimate

Higher orders capture more dependencies

Bigram Model

198015222 the first
194623024 the same
168504105 the following
158562063 the world
…
14112454 the door

23135851162 the *

Trigram Model

197302 close the window
191125 close the door
152500 close the gap
116451 close the thread
…
87298 close the deal

3785230 close the *

P(door | the) = 0.0006

P(door | close the) = 0.05

INCREASING N-GRAM ORDER

Unigram	• 'To him swallowed confess hear both. Which. Of save on trail for are ay device and rote life have • Every enter now severally so, let • Hill he late speaks; or! a more to leg less first you enter • Are where exeunt and sighs have rise excellency took of.. Sleep knave we. near; vile like
Bigram	• What means, sir. I confess she? then all sorts, he is trim, captain. •Why dost stand forth thy canopy, forsooth; he is this palpable hit the King Henry. Live king. Follow. •What we, hath got so she that I rest and sent to scold and nature bankrupt, nor the first gentleman? •Enter Menenius, if it so many good direction found'st thou art a strong upon command of fear not a liberal largess given away, Falstaff! Exeunt
Trigram	• Sweet prince, Falstafl shall die. Harry of Monmouth's grave. • This shall forbid it should be branded, if renown made it empty. • Indeed the duke; and had a very good friend. • Fly, and will rid me these news of price. Therefore the sadness of parting, as they say, 'tis done.
Quadrigram	• King Henry. What! I will go seek the traitor Gloucester. Exeunt some of the watch. A great banquet serv'd in; • Will you not tell me who I am? • It cannot be but so. • Indeed the short and the long. Marry. 'tis a noble Lepidus.